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Efficient Epileptic Seizure Prediction Using Single channel EEG Signal and Knowledge Distillation on Deep Neural Networks

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Abstract

Epilepsy is a common neurological disease, and in some patients, abnormal changes in brain activity typically begin before the onset of a seizure. Electroencephalography (EEG) is a practical method for recording electrical activity of brain and plays a crucial role in the diagnosis of epilepsy. Previous studies relied on multi-channel EEG signals and large deep neural networks, which require powerful hardware and reduce user convenience. In this study, teacher-student based knowledge distillation technique on deep neural networks is employed to find the best single EEG electrode to improve user convenience and decrease the network complexity. The teacher model employs Mel-spectrograms of all EEG electrodes as input of 3D convolutional layer of neural network. Subsequently, using response-based knowledge distillation techniques the information from 22 input electrodes of teacher model is transferred to the student model utilized one electrode. The proposed teacher and student models were evaluated on the CHB-MIT dataset, which contains scalp EEG signals recorded using 22 electrodes in a bipolar montage from 24 patients. The experimental results enabled us to identify two electrodes with superior performance (electrode no. 20 achieved accuracy of 84.49%, sensitivity of 86.85%, specificity of 82.58%, and an F1-score of 83.58%, while electrode no. 22 reached accuracy of 84.30%, sensitivity of 86.45%, specificity of 82.93%, and an F1-score of 83.35%). Both electrodes demonstrated the capability to predict seizure onset 30 minutes prior to its occurrence and have superior performance in terms of accuracy, sensitivity, specificity, and F1-score.

Keywords: Brain Signals, Deep Learning, Epilepsy, Knowledge Distillation, Seizure Prediction.

I. Introduction

Epilepsy is one of the most common neurological diseases in the world wide [1], affecting over 65 million people. Individuals with epilepsy are up to three times more likely to experience premature death compared to the general population [2]. Unfortunately, in 30% of patients, the condition remains uncontrolled despite medication. Therefore, research on epileptic seizure prediction (ESP) has particular importance, as it can significantly improve the quality of life for individuals with epilepsy. Patients with epilepsy frequently experience one or more seizures due to abnormal brain function. They are typically treated with medication, and in some cases, surgical intervention is employed. However, in more than 30% of cases, epileptic seizures cannot be controlled through surgery or medication [2]. An effective approach is to detect seizures a few minutes before their onset (a condition referred to as the preictal state) and manage them with medication [3]. Seizure prediction and supporting patients with epilepsy remains ongoing challenge in preventing seizure-related complications and improving the quality of life for those affected by this condition.

Electroencephalography (EEG) is a practical method for recording the brain's electrical activity and aids in the diagnosis of epilepsy. In recent years, numerous studies have

demonstrated that EEG signals can be utilized for seizure prediction [3-5]. Currently, the most advanced seizure prediction methods are based on the analysis of EEG signals using machine learning or deep learning techniques, focusing on identifying a seizure short period prior to onset it. Since the seizure occurrence structures vary significantly among patients, most existing methods (machine learning or deep learning models) are trained in a personalized manner (patient-specific models), meaning that a separate model is trained for each patient rather than a general model for all patients [5]. To this end, numerous studies have been conducted on seizure detection and prediction using EEG signals. Traditional machine learning methods are typically based on logistic regression, higher order statistics analysis [6], Support Vector Machine (SVM) [7] or integration of singular value decomposition (SVD) and graph theory [4]. However, these methods require prior knowledge to manually extract features from input data. Recently, deep learning-based approaches have been proposed to address the limitations of the time-consuming manually feature extraction process and its lack of generalizability. Among these approaches, Convolutional Neural Networks (CNNs) are used for automatic feature extraction, while Recurrent Neural Networks (RNNs) are employed to improve model temporal relationships [3]. The detection of seizures and the provision of effective support for individuals with epilepsy continue to pose major challenges in the effort to prevent seizure-related complications and improve the quality of life for those affected.

Electroencephalography is a non-invasive method that is placed on the scalp and causes no harm to the individual. Among non-invasive methods for examining human brain activity, electroencephalography provides a direct measurement of cortical activity with millisecond-level temporal resolution and uses an electrode cap to capture brain waves from the surface of the scalp [7]. Human brain waves are categorized into five distinct frequency bands: delta (1-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz), and gamma (30-100 Hz), each of which plays a specific role in brain activity [8]. The placement of EEG electrodes is typically configured using the international 10-20 system [9-10]. Figure 1 illustrates the placement of electrodes.

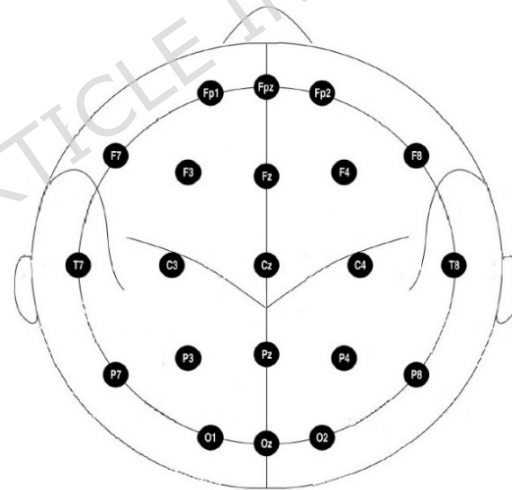


Figure 1. The International 10-20 System[10].

In general, long-term brain signals of individuals with epilepsy can be divided into four categories [11]:

- **Ictal:** This occurs when the patient experiences a seizure, causing changes in the brain's signals, which ends when the seizure begins and finishes.
- **Preictal:** Refers to the time before the onset of a seizure.
- **Postictal:** This state begins after the seizure has ended.
- **Interictal:** In this state, the patient is in a normal condition, with no signs of a seizure present.

Figure 2 shows four types of brain signals in individuals with epilepsy [12]. A common method for seizure prediction involves collecting surface or intracranial EEG signals, preprocessing, feature extraction, and classification between the interictal and preictal states. If the preictal state is successfully identified, an alert is generated to enable seizure control before its onset [4].

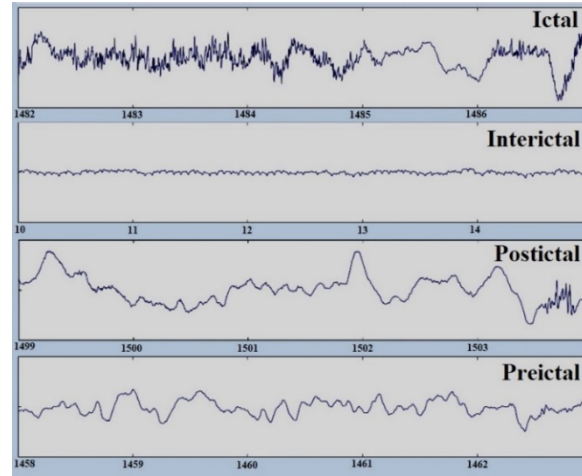


Figure 2. Different states of brain signals: ictal, interictal, postictal, and preictal [12].

One of the issues in previous works is the employing of multi-channel EEG with large neural networks, which requires powerful hardware, and needs to use the large number of electrodes, making it difficult for patients to work with it. The use of all channels has led to an increase in the computational load of deep models due to the number of electrodes and the volume of data. To have a practical system in which the patients can use it without needing to wear 22 electrodes on head and connect them to a system with high computational power, we must not only reduce the number of electrodes but also reduce the size of deep learning model. Although the methods such as feature selection techniques, channel-attention mechanisms, or mutual information-based feature ranking can be used for reducing the size of deep learning models, the number of electrodes remains a challenge that reduce convenience of patients and usability of the system. If one reduces the number of electrodes, the chance of employing the useful information from neglected electrodes is also lost. The knowledge distillation technique tries to use the useful information of all electrodes in a teacher model, and then transfer this information from teacher model to a lightweight student model. In this paper, knowledge distillation is employed to reduce the number of channels for user convenience and to decrease the network size.

Our goal is to implement a teacher-student-based system to improve accuracy and sensitivity for seizure prediction using deep learning methods. This research tries to reduce the complications [13] caused by seizures for the patient and assist caregivers in managing the patient's condition. Given that recent research on epilepsy seizure detection has increasingly emphasized machine learning approaches, particularly deep learning techniques and neural networks, which have shown substantial improvements over traditional methods. This study aims to propose a deep learning-based model for seizure prediction with enhanced accuracy. Another goal of this research is to predict seizure occurrence using a patient-independent model. Instead of designing and applying a separate model for each patient, a single model is used for all patients. This approach results in a model with higher generalizability. Also, the proposed method aims to reduce the number of EEG channels used for predicting the time of seizure onset. This is done by employing teacher-student based Knowledge Distillation (KD) method, where the student imitates the teacher's behavior. This model, which is a part of CNN models, consists of two components: the Teacher and Student networks. The Teacher model employs all EEG channels and tries to transfer its knowledge

to the light-weighted student model, which only uses one channel and also has a lower number of convolutional layers. Employing the knowledge distillation technique on a single electrode enables us to identify two electrodes with superior performance compared to the others through experimentation.

The remaining part of the paper is organized as follows: In the second section, we review the research conducted in this field. The third section discusses the proposed model, preprocessing, feature extraction, and knowledge distillation. The fourth section examines the experimental results, and the fifth section presents the conclusion.

II. RELATED WORK

In general, research on epileptic seizure prediction can be divided into two categories: patient-independent models and patient-specific models. A patient-specific model is trained using the data of a single patient and is also evaluated using that same patient's data. In contrast, a patient-independent model is trained on data from all patients and evaluated using data from the entire patient group. Existing studies are typically trained as patient-specific models, since epileptic signals often exhibit characteristics unique to each individual. Patient-independent model generally performs worse than personalized ones due to the high inter-patient variability [7]. In the following, we first review studies related to patient-specific models.

Usman et al. [2] employed a two Dimensional-Convolutional Neural Networks (2D-CNN) for feature extraction and used Long Short Term Memory (LSTM) networks for classification. They applied Empirical Mode Decomposition (EMD) to remove noise and addressed the class imbalance problem by generating synthetic data using a Generative Adversarial Networks (GAN). Daoud et al. [14] used raw EEG signals without any preprocessing as input to their model. They employed a 2D-CNN for both feature extraction and classification. Zhang et al. [15] employed the Phase Correlation Coefficient (PCC) method for feature extraction and utilized a CNN for classification. Meng et al. [16] employed mobile network information gain (M-NIG) EEG signals for epileptic seizure prediction. Xu et al. [17] addressed the issue of data imbalance by generating synthetic data using a GAN. Their model is a hybrid architecture combining CNN and RNN. Sharma et al. [18] employed a deep neural network for seizure classification. They also utilized higher order statistics for analyzing of EEG signals. Jana et al. [19] used raw EEG signals from three channels as input and employed a 1D-CNN for feature extraction and classification. Zhang et al. [20] employed a wavelet decomposition package for feature extraction, aiming to improve accuracy and reduce execution time. They employed a combination of a convolutional neural network, gated recurrent unit, and attention mechanism (CNN-GRU-AM) for classification. Gao et al. [21] utilized a CNN for both feature extraction and classification, and incorporated an attention mechanism to focus on more significant features. Liu et al. [22] employed a CNN for seizure prediction. Usman et al. [23] performed preprocessing using Short-Time Fourier Transform (STFT) and noise removal with a bandpass filter. They extracted features using statistical metrics and a 2D-CNN. For classification, they employed an ensemble approach combining three models: 2D-CNN, SVM, and LSTM modules. Ozcan et al. [24] evaluated three different approaches: Multilayer Perceptron (MLP), 2D-CNN, and 3D-CNN. Among these, the 3D-CNN model demonstrated the best performance. Usman et al. [25] utilized both bandpass and bandstop filters for noise removal and frequency band limitation. They further employed a CNN for feature extraction and an SVM for classification. Also, Sharma et al. [26] performed automated seizure classification using autoencoder based deep neural network.

Wei et al. [27] used a CNN-LSTM model with 10 EEG channels for seizure prediction, effectively capturing spatial and temporal features. Shahbazi et al. [28] applied STFT for preprocessing and used a combination of 2D-CNN and LSTM for classification. Georgis et al. [29] used STFT on input EEG and performed seizure prediction using both supervised and unsupervised approaches. They employed three networks: CNN, LSTM and Temporal Convolutional Network. Yang et al. [30] performed seizure prediction using both EEG and ECG signals. Feature extraction from EEG signal was carried out using 1D-CNN, and LSTM

was employed for classification. Aslam et al. [31] employed CNN and LSTM for feature extraction and classification in their seizure prediction model. Osman et al. [6] performed preprocessing using Empirical Mode Decomposition (EMD), bandpass filter, and STFT. They increased the data quantity using GAN. For classification, they employed a combination of LSTM, CNN, and SVM in the MAML framework.

Zhao et al. [32] preprocessed the raw data using the STFT and computed the spectrogram, which was then used as input to the model. The model they employed was a combination of Gated Recurrent Unit (GRU) and LSTM. Pandey et al. [33] combined two optimizers, Forensic-based-Search-and-Rescue Optimization (FB-SARO), with LSTM and used it as the model. Ryu et al. [34] initially transformed the EEG signal into the time-frequency domain using the discrete wavelet transform and then used it as input to a combined DenseNet and LSTM networks. Tsiouris et al. [35] performed seizure prediction using LSTM model without noise removal, and by employing only raw EEG data.

Lu et al. [36] used the STFT for preprocessing. They employed a 3D-CNN and Bi-LSTM for feature extraction and classification, while Bajaj et al. [35] similarly used raw data with a CNN-Bi-LSTM for automatic feature extraction and classification.

Liu et al. [37] used the radial basis function neural network method to predict seizure status using 18 channels common among the patients. Cui et al. [38] used ELM for classification. Rasheed et al. [39] employed Inceptionv3 model using transfer learning. Yang et al. [40] used the spectrogram as input. They also employed the ResNet network as the model and further applied an attention mechanism. Liu et al. [41] performed feature extraction using power spectral density. Peng et al. [42] used frequency-domain for feature extraction.

The following section examines patient-independent models. The patient-independent models offer a more practical solution for seizure prediction. The number of studies employing patient-independent models remains relatively limited; the following section provides a review of these approaches. Khan et al. [43] developed a patient-independent model using 2D-CNN for seizure prediction. In this study, the wavelet transform was employed for preprocessing and used as the input to the trained 2D-CNN which consists of six convolutional layers. The output layer contains three units with a softmax activation function, representing the probability distribution across three classes: ictal, preictal, and interictal. Wu et al. [44] selected 11 patients with primary seizures and identical channel counts and arrangements. They designed a CNN-based model inspired by the ResNet architecture. Dissanayake et al. [45] segmented the preictal class with a 50% overlap, while the interictal class was segmented without any overlap. They employed Mel-frequency cepstral coefficients (MFCC) for feature extraction, with dimensions (23, 13, 201), where 23 corresponds to the number of EEG channels, 13 represents the number of MFCC features, and 201 indicates the number of time windows. Additionally, they incorporated graph-based representations of the EEG signals. Their model architecture consisted of two-dimensional convolutional layers, followed by a LSTM network. The number of convolutional layers used in their model is 3, and they also employed three LSTM layers. They also used a 2D-CNN for classification [46].

Hu et al. [47] designed a seizure prediction algorithm for EEG signals. The amplitude spectra of EEG signals from 18 channels were first computed and then divided into 19 frequency sub-bands. The mean amplitude spectrum (MAS) in each of 19 frequency sub-bands was calculated for every channel, resulting in a MAS map with dimensions of 18×19 . Finally, this MAS map was fed into a convolutional neural network (CNN) for feature extraction, and an SVM was used for seizure prediction. They employed two 2D convolutional layers with 128 and 256 filters, respectively, each using a kernel size of (5,5). Each convolutional layer was followed by a max-pooling layer with a window size of (2,2). Cho et al. [48] investigated the use of Phase Locking Value (PLV) for seizure prediction, particularly focusing on the gamma frequency band. They utilized both simulated data and 65 clinical seizure cases for their study. In addition, various filtering algorithms were employed to decompose the spectral components of the data, including bandpass filtering, Empirical Mode Decomposition (EMD), Multivariate Empirical Mode Decomposition (MEMD), and Noise-Assisted Multivariate Empirical Mode Decomposition (NA-MEMD). They used an SVM for classification.

Ultimately, the gamma frequency band proved to be effective in distinguishing between the preictal stage and the interictal state. Peng et al. [49] employed STFT and power line noise removal for preprocessing, subsequently developed a seizure prediction model using Maximum Mean Discrepancy Adversarial Auto encoder(MMD-AAE). Building upon their previous work, Peng et al. [50] again employed STFT and power line noise removal for preprocessing, and utilized Manifold Adversarial Auto encoder(MAAE) as the seizure prediction model.

In general, patient-specific models are trained separately for each individual, typically requiring large neural networks. As a result, the training process must be repeated for each patient, and the resulting models are highly specific to the data of the corresponding patient. In the patient-independent model, a large number of electrodes have been used for each patient. Overall, both existing approaches rely on models with a large number of parameters, which are not suitable for deployment on low-power devices. Additionally, they require numerous electrodes on the patient's scalp, which not only increases the model complexity but also causes inconvenience for the user. In the proposed method, we aim to predict seizures by designing a lightweight model with a reduced number of parameters and using only single EEG electrode. Another key innovation in this research is the employing of teacher-student Knowledge Distillation (KD). The teacher model, which employs five layers CNN, is trained on the full 22-electrodes EEG data, capturing comprehensive brain activity information. This model gains a deep understanding of the EEG signals. The student model is only composed of four simple layers CNN with only one electrode. The teacher model, with more layers and electrodes, is capable of capturing more intricate features from the 22-electrodes EEG signals. In contrast, the student model, with fewer CNN layers and kernels, can independently extract key local features from one electrode. Nevertheless, the KD technique serves as a bridge, enabling the simpler student network to learn the valuable and rich knowledge of the teacher model. By transferring this knowledge, the student model is able to capture essential features with remarkable efficiency, ultimately achieving performance comparable to that of the more complex teacher model. This approach not only amplifies the student model's capacity to extract critical information but also preserves its computational efficiency, offering a robust yet efficient solution for EEG signal analysis. This transfer creates a lighter, more computationally efficient model, while still maintaining a high level of accuracy, showcasing the power of knowledge distillation in preserving performance with reduced computational requirements. This approach not only makes the system suitable for implementation on low-resource devices but also improves user convenience by eliminating the discomfort of multiple electrodes.

III. METHODOLOGY

In the proposed method, the teacher model is first designed with the goal of seizure prediction using all electrodes with high accuracy. Subsequently, the student model is constructed by compressing the teacher model. Through the application of knowledge distillation, the student model learns from the knowledge of the teacher. The architecture of the teacher and student models is illustrated in Figures 3 and 4, respectively.

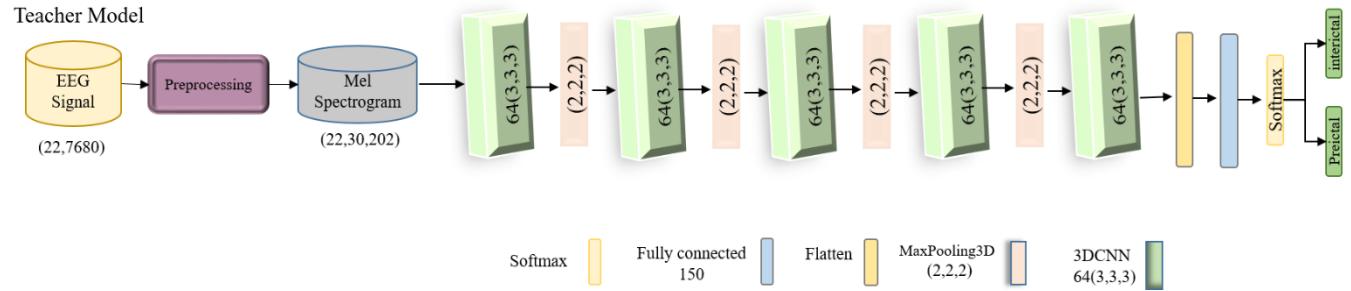


Figure 3. The Proposed Teacher Model

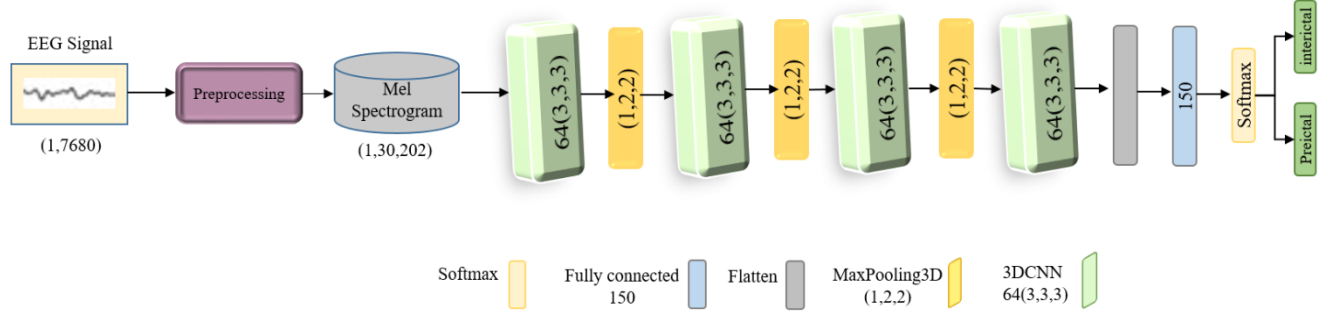


Figure 4. The Proposed Student Model

A. Preprocessing and Mel Spectrogram Feature Extraction

In both proposed teacher and student models, preprocessing is performed using two third-order Butterworth filters: a band-pass filter and a band-stop filter. The band-stop filter is used to remove power line noise in the frequency range of 50 to 60 Hz, effectively eliminating signals within this frequency range [6, 25, 39, 51]. The sampling frequency, as specified by the dataset, is 256 Hz. A band-pass filter is applied to retain frequency components in the range of 0.1 to 100 Hz. Researchers suggested that for epileptic seizure detection, the frequency band 0.1 to 70 Hz is generally more suitable, as it contains more relevant information [30]. Additionally, Z-score normalization is employed to standardize the input data.

Also, we employ the Mel-spectrogram [52] for feature extraction, resulting in dimensions of (22, 30, 202), where 22 represents the number of channels, 30 is the number of Mel bands, and 202 denotes the number of time windows used in each Mel-spectrogram. Using Mel scale scenario, the EEG frequency range 0.1 to 70 Hz is logarithmically spanned by 30 Mel triangular filter bank. Then the Mel Spectrogram is calculated for each electrode signal x . To calculate the Mel spectrogram, first, the STFT is applied to the electrode's signal x according as Equation (1):

$$X(n, \omega) = \sum_{m=-\infty}^{\infty} x[m] \cdot \omega[n - m] \cdot e^{-i\omega m} \quad (1)$$

where $x[m]$ is input EEG channel signal, $\omega[n - m]$ is Hamming windowing function, $X(n, \omega)$ is Time-Frequency representation of the signal. Then, the power spectrogram is computed according to Equation (2):

$$P(n, \omega) = |X(n, \omega)|^2 \quad (2)$$

Then Mel filter bank is applied according to the formula provided in Equation 3.

$$S(m, n) = \sum_{k=1}^K P(n, k) \cdot H_m(k) \quad (3)$$

$H_m(k)$ is the Mel filter function in a triangular shape. $S(m, n)$ denotes the energy of the m^{th} Mel band at time n . The calculation of the Mel energy logarithm (Log-Mel Spectrogram) is performed according to Equation 4.

$$\text{MelSpec}(m, n) = \log(S(m, n) + \epsilon) \quad (4)$$

In Equation 4, ϵ is a very small value that prevents the logarithm of zero. Figure 5 shows an example of preprocessing and Mel-spectrogram extraction.

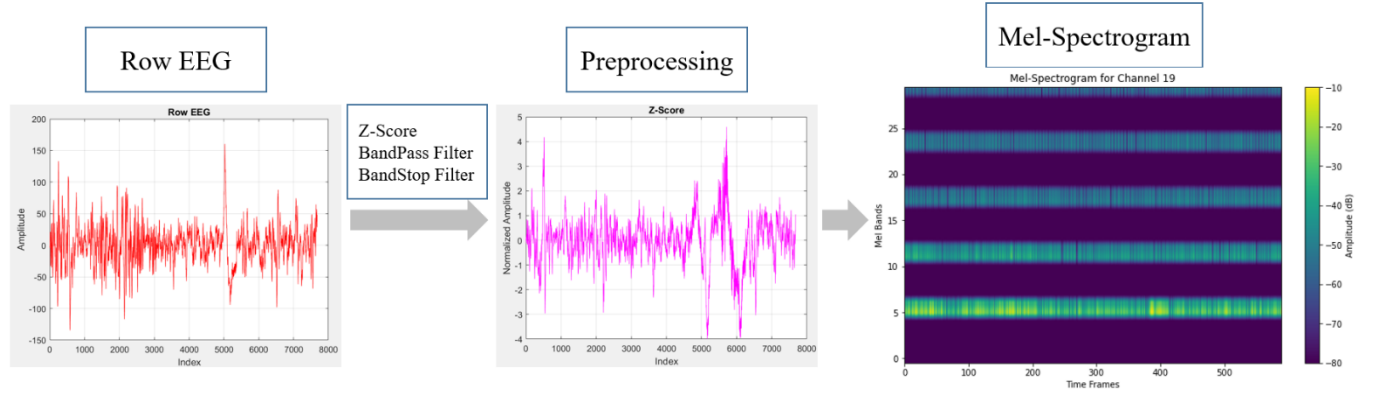


Figure 5. An example of preprocessing and Mel-spectrogram output for channel 19 in the preictal state.

B. Teacher Model

The input to the teacher model is the Mel-spectrogram of 22 EEG channels. This model employs three-dimensional convolutional layers, each with 64 filters and a uniform kernel size of (3, 3, 3) across all convolutional layers. After each four convolutional layers, known as the Teacher 3D-CNN block, a MaxPooling layer with a kernel size of (2, 2, 2) is applied. The final convolutional layer does not include a MaxPooling layer. The ReLU activation function is used in each convolutional layer while preserving the input dimensions. After each convolutional layer, Dropout with a rate of 0.5 is applied to deactivate a portion of the neurons, and MaxNorm with a constraint of 0.4 is employed to regulate the weights and prevent overfitting. In the final layer, the Softmax activation function is used, and the loss function is categorical cross-entropy. This model is employed as the teacher model for knowledge distillation framework. After training the teacher model, an effective and advanced knowledge distillation technique is utilized to transfer the knowledge from the teacher model to the student model, enabling the optimized model to maintain high accuracy and performance while being suitable for real-time applications especially on resource-constrained devices. The primary objective of this process is to achieve a balance between reducing computational complexity and maintaining prediction ability.

C. Student Model

The student model is designed as a smaller version of the teacher model, maintaining a similar structure while reducing the network's size. The student model consists of four 3D-CNN blocks and three MaxPooling layers with equal kernel sizes. Unlike the teacher model, the student model performs seizure prediction using only a single EEG channel instead of 22 channels. To find the best channel, we evaluate the performance of each channel and the one with the highest accuracy is selected for final student model. Similar to the teacher model, the student model applies a MaxPooling layer with the same kernel size after each of the first three convolutional layers, while the final convolutional layer does not include a MaxPooling layer. The kernel size of the MaxPooling layers used in the student network is (1, 2, 2) for all three layers. The ReLU activation function is applied in each convolutional layer while preserving the input dimensions. After each convolutional layer, known as Student 3D-CNN block, a Dropout rate of 0.5 is used to deactivate a portion of neurons, and MaxNorm regularization with a threshold of 0.4 is employed to control the weights and prevent overfitting. Moreover, the loss function used is categorical cross-entropy. This model is employed as the student model in the knowledge distillation framework. After designing the student network, we adopted a response-based knowledge distillation approach, enabling

the student model to improve its performance by learning from input training data and from the knowledge transferred by the teacher network.

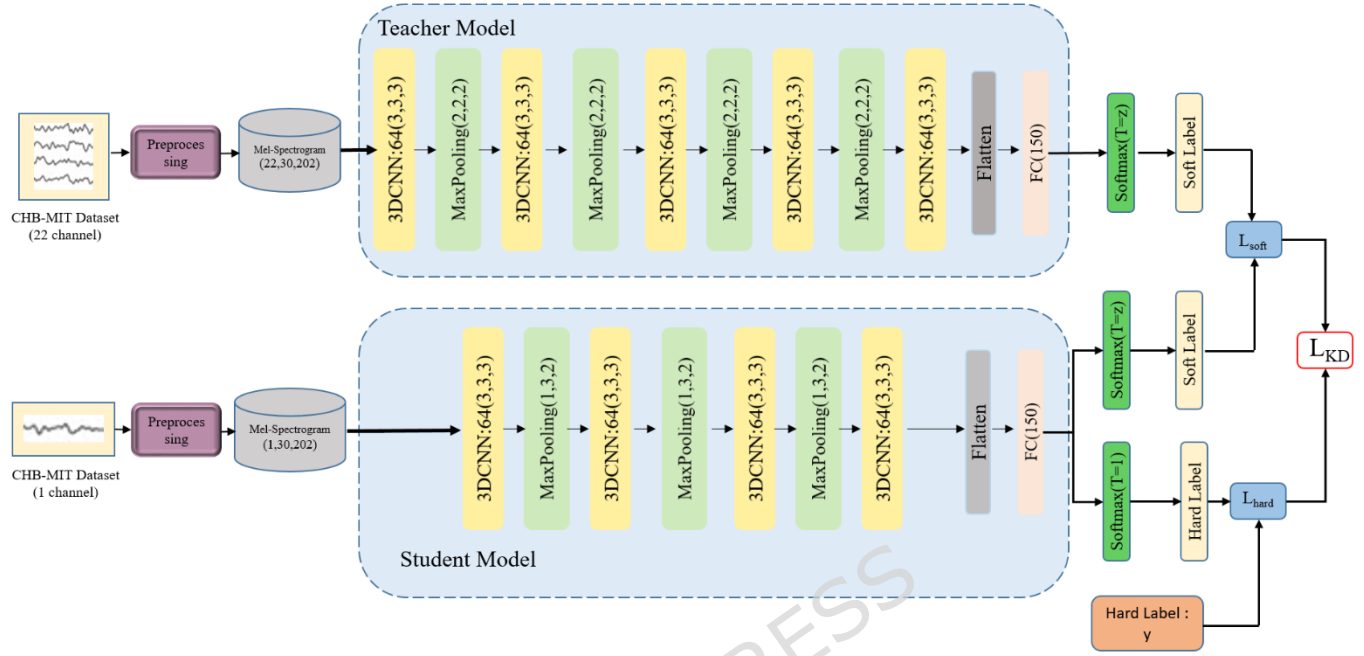


Figure 6. The Proposed Knowledge Distillation Model.

D. Response-based Knowledge Distillation

In the response-based approach, as shown in Figure 6, the loss function of the model output is computed by combining the loss functions of both the teacher and the student models. Knowledge distillation simultaneously leverages both hard labels and soft labels. The idea is that the teacher model contains richer information about uncertainties—such as class similarities—in its output, and this information can be effectively transferred to the student model.

A hard label refers to the actual ground-truth label of the data, typically denoted by y . z_i is the output of the teacher model before applying the softmax function, and used for computing the teacher's soft labels, represented by q . k represents the number of classes.

$$q = \{q_i\}_{i=1}^k \quad (5)$$

q_i is calculated using Equation (6):

$$q_i = \frac{\exp(\frac{z_i}{T})}{\sum_{i=1}^k \exp(\frac{z_i}{T})} \quad (6)$$

The variable T is used for smoothing the probability distribution output and plays a crucial role in knowledge transfer. For hard labels, the value of T is set to one, whereas for calculating soft labels, it is considered to be greater than one [53]. The student model is trained using both soft and hard labels. Then, using the same method, the soft labels for the student model are also computed. Here, s_i represents the output of the student model before the softmax, and p denotes the soft labels of the student, which are calculated according to Equation (7):

$$p_i = \frac{\exp(\frac{s_i}{T})}{\sum_{i=1}^k \exp(\frac{s_i}{T})} \quad (7)$$

Next, to calculate the loss function for soft labels, The Cross-entropy between the soft labels of the teacher model and the soft labels of the student model is computed according to Equation (8):

$$L_{\text{Soft}} = - \sum_{i=1}^k q_i \log p_i \quad (8)$$

Then, the loss function for hard labels is also computed using The Cross-entropy between the true labels y and the soft labels of the student model p , according to Equation (9):

$$L_{\text{Hard}} = - \sum_{i=1}^k y_i \log p_i \quad (9)$$

The objective of response-based knowledge distillation is to directly transfer the final layer's output from the teacher model to the student model, so that the student can mimic the teacher's final predictions. Ultimately, the knowledge distillation loss can be computed using both the hard and soft loss functions, as shown in Equation (10). A hyperparameter α is employed to assign a weight to each of the soft and hard loss components [54, 55].

$$L_{\text{KD}} = \sum_{i=1}^n ((T^2 \alpha) L_{\text{Soft}} + (1 - \alpha) L_{\text{Hard}}) \quad (10)$$

Accordingly, the teacher model is trained using 22 EEG channels. While the student model is implemented as a smaller network compared to the teacher and is trained using data from a single EEG channel. In the knowledge distillation model (as shown in Figure 6), the student attempts to mimic the behavior of the teacher. This is done to achieve performance comparable to the teacher model while using a more lightweight architecture. In the knowledge distillation approach, the teacher model is significantly more complex and advanced than the student model, requiring input from all EEG channels and demanding substantial computational resources for execution.

IV. RESULTS

In this study, the CHB-MIT dataset [12], which was collected at the Children's Hospital Boston in collaboration with the Massachusetts Institute of Technology (MIT) was used for evaluating the proposed models. The dataset is publicly available on the PhysioNet platform [12]. This dataset includes 24 scalp EEG recordings from patients diagnosed with epilepsy. The EEG signals were recorded using the standard 10-20 electrode placement system. These signals were sampled at a rate of 256 Hz with 16-bit resolution, and in most sessions, 23 electrodes were used [2, 56]. This dataset includes recordings from 23 epilepsy patients (5 males, aged 3 to 22 years, and 17 females, aged 1.5 to 19 years). Patient number 21 is actually the same as patient number 01, with EEG signals recorded approximately 1.5 years later. Overall, the dataset contains approximately 644 hours of EEG recordings, most of which are segmented into one-hour sessions [56]. Patient characteristics are summarized in Table 1. Each recording includes a variable number of electrodes—ranging from 18 to 32—depending on the patient, although most share a common set of 23 electrodes. In this study, a total of 180 usable seizures were recorded using 23 bipolar EEG channels. Upon further examination, it was evident that channel 23 in the dataset was a repetition of channel 15. Therefore, utilized channel 23 was removed, and 22 bipolar montage channels were used in this study. The data were segmented into 30-second frames. For class 0 (normal state), the frames were prepared without overlap, while for class 1 (preictal state), a 50% overlap was applied. Then, Mel-spectrograms with dimensions (22, 30, 202) were extracted from each sample. We set the preictal window to 30 minutes as used in the prior literature such as Wu et al. [44], Zhange et al. [57], Usman [2,23], Yang.2021[40], Gao [21], Pandey [33], Peng [49,50]. Additionally, the sampling rate is 256 Hz, and the data has been band-pass filtered within the frequency range of 0.1 to 100 Hz. The Mel-spectrogram values are defined as follows: 22 represents the number of channels, 30 is the number of Mel bands, and 202 is the number of time frames. The total number of data samples is 33,000 for both classes 0 and 1 combined. The number of samples for class 1 is 15,000, extracted from 141 preictal files using a 50% overlap. For class 0, there are 18,000 samples extracted from 180 normal (Interictal) files without overlap.

Table 1. Specifications of the CHB-MIT Dataset

patient ID	Gender	Age	Seizure No ¹	Seizure file	Non-seizure
Chb01	F	11	7	7	42
Chb02	M	11	3	3	36
Chb03	F	14	7	7	38
Chb04	M	22	4	3	42
Chb05	F	7	5	5	39
Chb06	F	1.5	10	7	18
Chb07	F	14.5	3	3	19
Chb08	M	3.5	5	5	20
Chb09	F	10	4	3	19
Chb10	M	3	7	7	25
Chb11	F	12	3	3	35
Chb12	F	2	27	13	24
Chb13	F	3	10	8	33
Chb14	F	9	8	7	26
Chb15	M	16	20	14	40
Chb16	F	7	8	6	19
Chb17	F	12	3	3	21
Chb18	F	18	6	6	36
Chb19	F	19	3	3	30
Chb20	F	6	8	6	29
Chb21	F	13	4	4	33
Chb22	F	9	3	3	31
Chb23	F	6	7	3	9
Chb24 ²	-	-	15	12	22
Total	-	-	180	141	686

To evaluate the proposed model in the patient-independent scenario, instead of using the leave-one-patient-out technique, a 5-fold cross-validation was used (in each run 72% of the data was used for training, 8% for validation, and 20% (one fold) for testing). To maintain independency, we first manually divided 24 patients into five folds. Then all samples of each patient were placed in the same fold. Therefore, the number of samples of all folds is approximately equal.

To evaluate the proposed models, accuracy, precision, sensitivity, specificity, F1-score, and FPR were used. The following equations represent the calculations of accuracy, sensitivity, specificity, F1-score, and false positive rate (FPR), respectively.

¹ Some seizure files contain multiple seizures within a single file.

² In the dataset description, no information is provided about patient number 24.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FN+FP} \quad (11)$$

$$\text{Sensitivity} = \frac{TP}{TP+FN} \quad (12)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (13)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (14)$$

$$\text{F1 - Score} = 2 \times \frac{\text{prec} \times \text{sen}}{\text{prec} + \text{sen}} \quad (15)$$

$$\text{FPR} = \frac{\text{Number of False Alarms}}{\text{Total Recording Time(hours)}} \quad (16)$$

In these equations, the four values represent true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Sensitivity must be calculated to ensure that preictal data correctly classified as seizures. Moreover, specificity is considered as an important metric because if interictal states are mistakenly identified as preictal, unnecessary administration of preventive medication could potentially harm the patient's health. Moreover, prediction time is considered one of the key performance metrics. FPR, which stands for False Prediction Rate, is defined as the number of false alarms per hour. Methods that provide longer seizure prediction horizons offer greater opportunity for medical intervention, enabling patients and their caregivers to take preventive measures before the onset of a seizure. In this study, the preictal interval length (PIL) is considered to be 30 min before the onset of each seizure. The seizure prediction horizon (SPH) is also defined as 30 min, so that a correct prediction must be made at most 30 min before the onset of the seizure. The seizure occurrence period (SOP) is the zero min interval after seizure prediction time [44]. Therefore, any correct diagnosis of the preictal class in SPH that leads to the occurrence of a seizure is considered as a successful prediction, and this time is carefully controlled and checked for each seizure event.

Table 2 presents the initial parameters used for implementing the models. Multiple experiments for the proposed models were conducted on hardware equipped with an Intel i9-10900 CPU, 128GB RAM, and an NVIDIA-GeForce-RTX-3090 GPU with 24GB memory. The proposed model was implemented in the Anaconda environment using Python version 3.10 and executed on the TensorFlow framework with the Spyder IDE.

Table 2. Initial Parameter Values

Parameters	Value
Frame Size	30 s
Prediction time	30 min
Epochs	250
Learning Rate	0.0001
Optimizer	Adam
Batch Size	32
Dropout	0.5
α	0.9
T	2

A. Teacher Model Evaluation

The extracted data was divided into five folds and evaluated accordingly. Four of these folds were used for training and validation, while the remaining one was used for testing. Finally, the Average and variance of the results were calculated. The outcomes were presented in Table 3. According to Table 3, it can be observed that the five folds evaluated in the experiments demonstrated relatively similar performance, yielding results that are closely aligned across all evaluation metrics.

Figure 7 illustrates the accuracy and loss curves for both training and validation phases of the teacher model across fold 5. The curves indicate that the training and validation results

were relatively similar. Figure 8 presents the confusion matrices of the teacher model for the five evaluated folds. Based on the figure, it can be observed that the teacher model exhibited relatively similar performance across all five folds.

Additionally, for the teacher model, several parameters—such as the number of Mel bands, dropout rate, and number of filters—were explored and tested to achieve better performance. The results of these experiments are presented in Figure 9. As shown in Figure 9, based on the experiments conducted on the teacher network, part (a) examines the number of Mel bands. The model achieved better performance with 30 Mel bands, while a decrease in accuracy was observed for other values. In Figure 9(b), the dropout rate was tested and analyzed, and as a result, a dropout value of 0.5 demonstrated better performance compared to the other tested values. Finally, in Figure 9(c), which corresponds to the number of kernel filters in each convolutional layer, the model's performance was evaluated based on the number of filters. It was observed that using 64 filters resulted in better performance compared to the other tested values.

B. Student Model Evaluation

The student model was initially evaluated independently to assess its standalone performance without the use of knowledge distillation. Thereafter, knowledge distillation was introduced as a means to enhance the model's overall performance. Table 4 presents the results of the single-channel student model both without and with employing of knowledge distillation. According to Table 4, it can be observed that the Mel-spectrograms of approximately the first 12 electrodes contain limited information for seizure prediction. These electrodes correspond to brain regions that are located farther from the temporal lobe. Based on Table 4, after applying knowledge distillation, electrodes 21, 20, and 22 performed better than the other electrodes across all metrics. Considering that the three electrodes had similar performance, and for the convenience of the user with two electrodes placed exactly behind the ear, electrodes 20 and 22 were selected. The results of the knowledge distillation model for electrodes 20 and 22 are provided in Tables 5 and 6 for 5-fold evaluation.

In Figure 10, the training and validation accuracy and loss curves, along with the confusion matrices for the knowledge distillation model corresponding to channels 20 and 22, are illustrated. According to the plots, the training and validation performances were relatively similar.

Also, appropriate values for the parameters T and α in the knowledge distillation model were evaluated and tested, and the experimental results were presented in Figure 11. In Figure 11 (a), values of T ranging from 1.1 to 4 were tested, and it was observed that $T = 2$ yielded better performance in the knowledge distillation model. Additionally, to determine an appropriate value for α , values from 0.1 to 1 were evaluated. As shown in Figure (b), $\alpha = 0.9$ it demonstrated superior performance.

Table 3. Results of Teacher Model Experiments Using 5-Fold Cross-Validation

Fold Number	Accuracy	Sensitivity	Specificity	F1-Score	FPR	Train Data	Test Data	Train time(s)	Test time(s)
Fold 1	87.61%	88.04%	87.23%	86.86%	0.1276	26421	6580	16124.36	5
Fold 2	87.82%	90.65%	85.54%	86.88%	0.1445	26408	6593	16192.33	5
Fold 3	87.27%	88.12%	86.56%	86.32%	0.1343	26384	6617	16149.92	7
Fold 4	86.95%	86.78%	87.10%	85.97%	0.1289	26400	6601	16255.25	5
Fold 5	87.67%	90.51%	85.38%	86.76%	0.1461	26388	6613	16240.44	5
Average	87.52%	88.82%	86.36%	86.56%	0.1362	-	-	16192.46	5.4
Variance	0.1231	2.8663	0.7443	0.1595	7.4432e-05	-	-	-	-

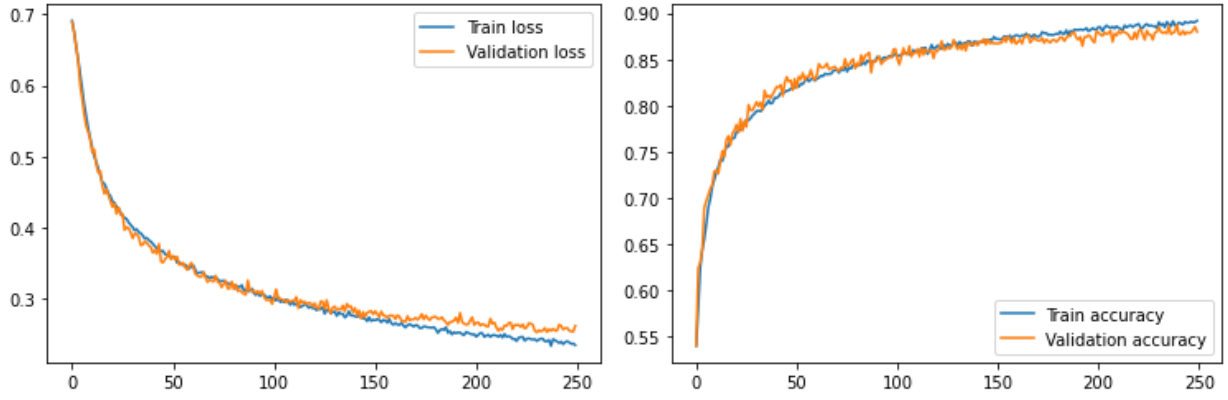


Figure 7. Training curves of the teacher network for accuracy and validation in fold-5: (a) Loss function, (b) Accuracy for fold-5.

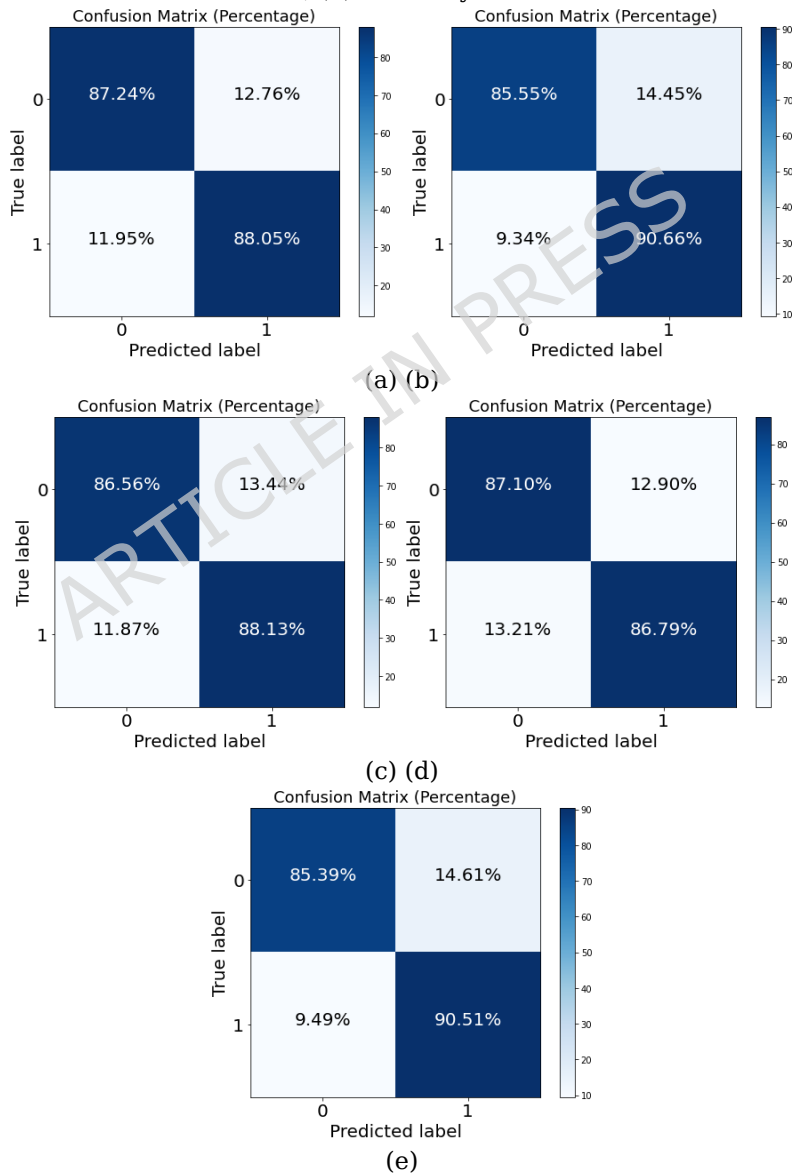
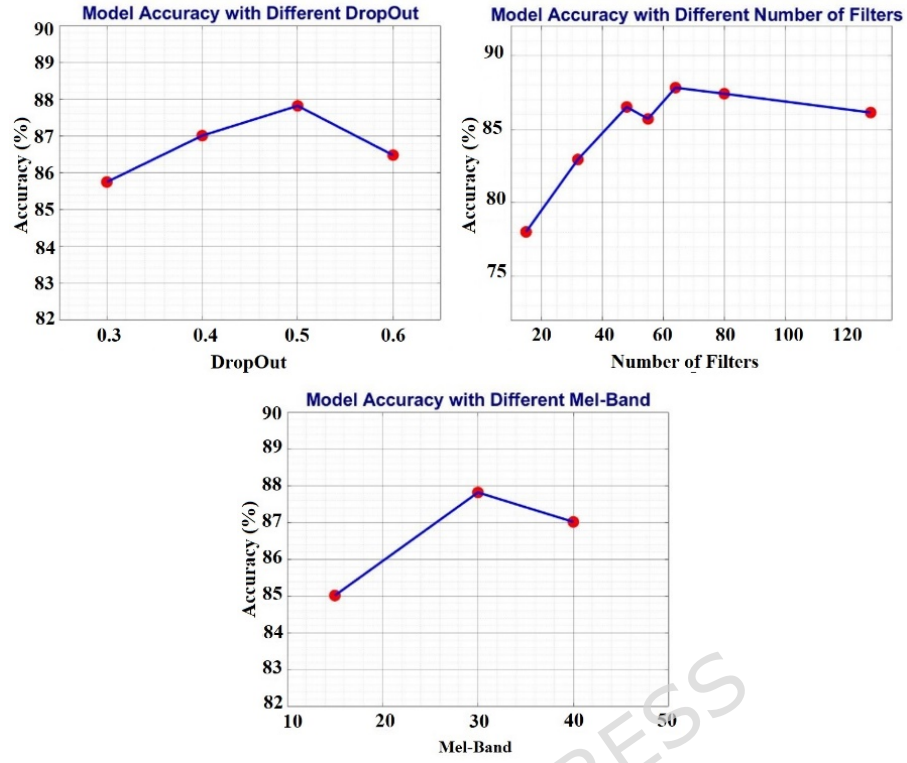
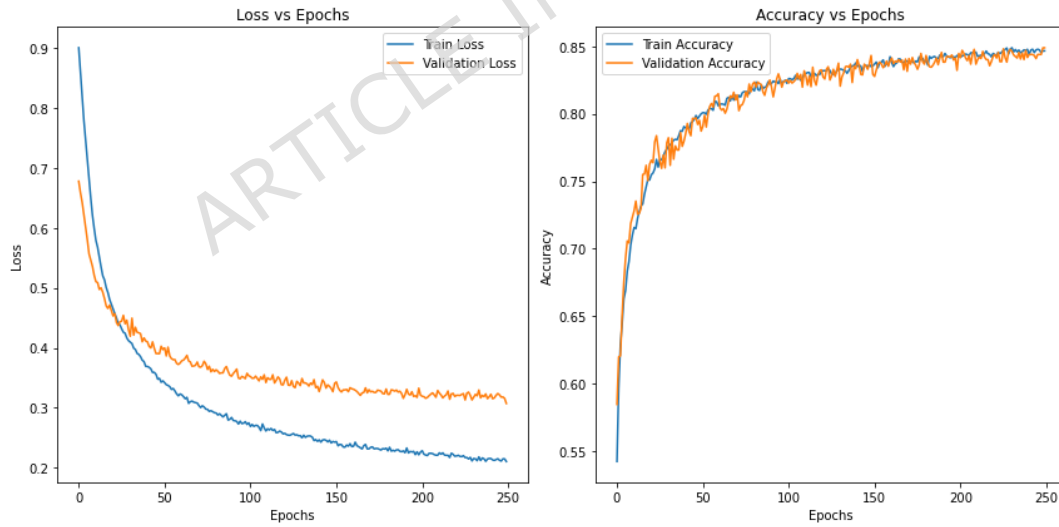


Figure 8. Confusion matrices of the teacher model: (a) Fold 1, (b) Fold 2, (c) Fold 3, (d) Fold 4, (e) Fold 5.



(a) (b) (c)

Figure 9. Evaluation of the impact of teacher model parameters: (a) number of Mel bands, (b) dropout rate, and (c) number of filters in 3D convolution.



(a)

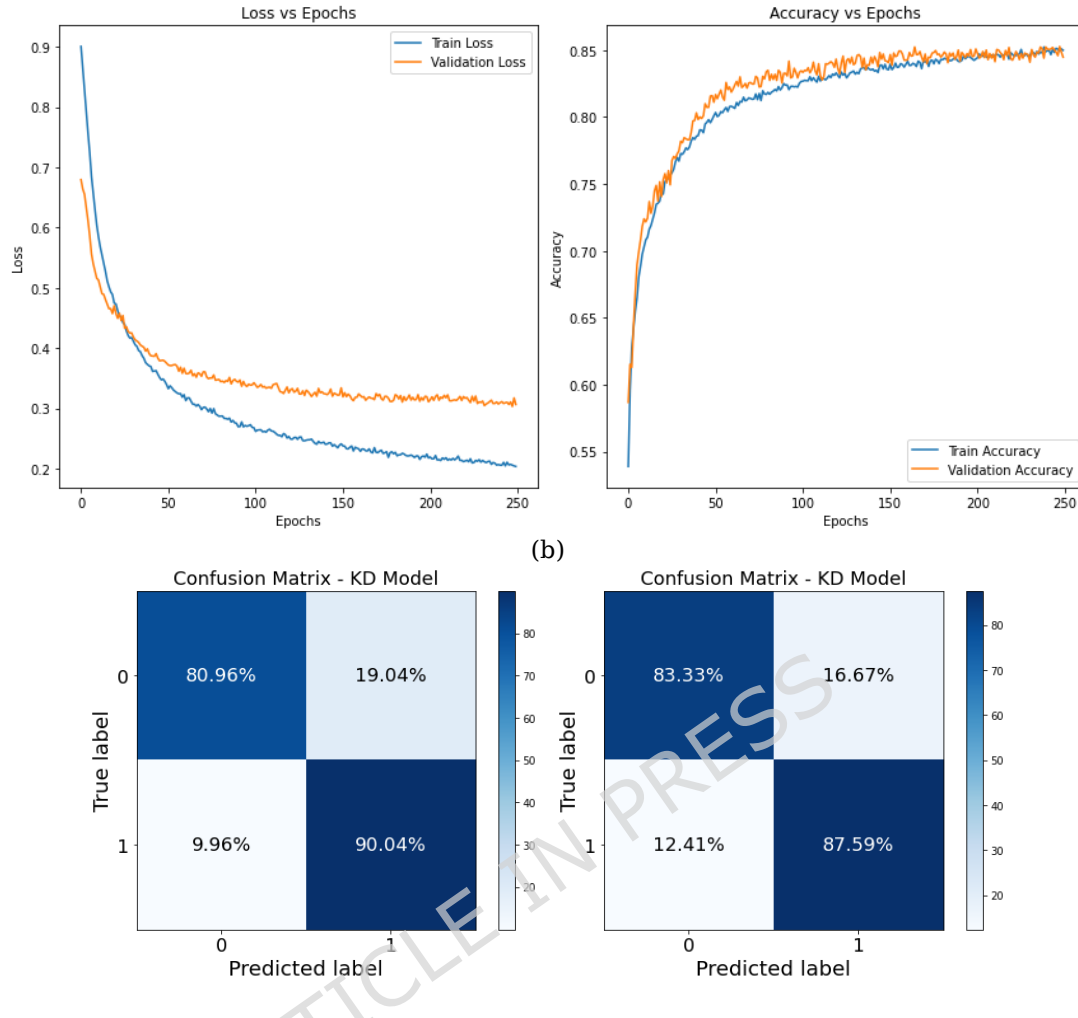


Figure 10. Training and validation plots, and confusion matrices: (a) Accuracy and loss of the knowledge distillation model for channel 20, (b) Accuracy and loss of the knowledge distillation model for channel 22, (c) Confusion matrix of the knowledge distillation model for channel 20, (d) Confusion matrix of the knowledge distillation model for channel 22.

Table 4. Results of the student model (single-channel) with and without applying knowledge distillation

Student model		without knowledge distillation					with knowledge distillation				
Row No	Channel name	Accuracy	Sensitivity	Specificity	F1	FPR	Accuracy	Sensitivity	Specificity	F1	FPR
1	FP1-F7	55.51%	0	100%	0	0	55.51%	0	100%	0	0
2	F7-T7	55.51%	0	100%	0	0	55.51%	0	100%	0	0
3	T7-P7	55.51%	0	100%	0	0	55.51%	0	100%	0	0
4	P7-O1	55.51%	0	100%	0	0	55.51%	0	100%	0	0
5	FP1-F3	55.51%	0	100%	0	0	55.51%	0	100%	0	0
6	F3-C3	55.51%	0	100%	0	0	55.51%	0	100%	0	0
7	C3-P3	55.51%	0	100%	0	0	55.51%	0	100%	0	0
8	P3-O1	55.51%	0	100%	0	0	55.51%	0	100%	0	0
9	FP2-F4	55.51%	0	100%	0	0	55.51%	0	100%	0	0
10	F4-C4	47.17%	68.53%	30.05%	53.57%	0.6994	55.68%	1.2%	99.29%	2.4%	0.0071

11	C4-P4	54.20%	6.1%	92.75%	10.60%	0.0724	56.97%	13.57%	91.75%	21.91%	0.2191
12	P4-O2	56.08%	5.7%	96.44%	10.39%	0.0355	57.91%	26.42%	83.14%	35.84%	0.1686
13	FP2-F8	47.11%	69.11%	29.48%	53.57%	0.7051	63.96%	53.05%	72.70%	56.71%	0.2730
14	F8-T8	54.96%	57.92%	52.59%	53.36%	0.4740	70.85%	64.61%	75.85%	66.35%	0.2415
15	T8-P8	58.60%	45.0%	69.50%	49.17%	0.3049	78.19%	78.93%	77.60%	76.30%	0.2240
16	P8-O2	63.76%	55.74%	70.19%	57.78%	0.2980	81.71%	82.58%	81.01%	80.07%	0.1899
17	FZ-CZ	70.46%	75.21%	66.66%	69.38%	0.3333	82.78%	85.0%	81.01%	81.46%	0.1899
18	CZ-PZ	82.37%	86.70%	78.90%	81.40%	0.2109	83.57%	85.48%	82.05%	82.24%	0.1795
19	P7-T7	84.06%	86.95%	81.63%	83.20%	0.1835	83.98%	86.87%	81.59%	81.14%	0.1841
20	T7-FT9	84.18%	83.60%	84.64%	82.46%	0.1535	84.49%	86.85%	82.58%	83.58%	0.1741
21	FT9-FT10	81.92%	83.12%	80.95%	80.35%	0.1904	84.48%	88.08%	81.51%	83.76%	0.1848
22	FT10-T8	82.29%	84.65%	80.40%	80.97%	0.1959	84.36%	86.45%	82.65%	83.40%	0.1735

Table 7 presents a comparison of the number of parameters between the teacher and student models. As one can be observed, the student network, with a reduced number of parameters, results in shorter training and testing times. The values presented in Table 7 correspond to training on four data folds and testing on one fold. In Table 7, for example, the testing time reported for one fold corresponds to the evaluation of all 6,580 data samples at once. As shown in Figure 12, these two electrodes are located in the temporal lobe of the brain. Electrode No. 20, named T7-FT9 in a bipolar montage, is positioned on the left side of the head, near the ear, within the temporal lobe. Similarly, Electrode No. 22, named FT10-T8, is situated on the right side of the head in the temporal lobe. These two electrodes demonstrated superior performance compared to the other electrodes used in the dataset. Finally, the comparison of the proposed method with existing methods presented in Table 8, related works with evaluation methods similar to those in this paper are compared. As can be seen, researchers have predicted seizures using various methods.

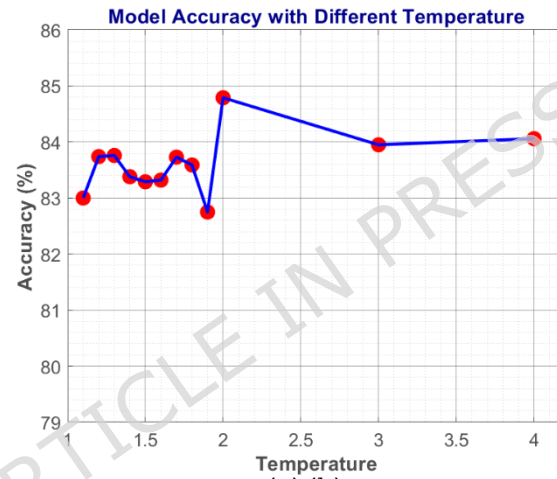
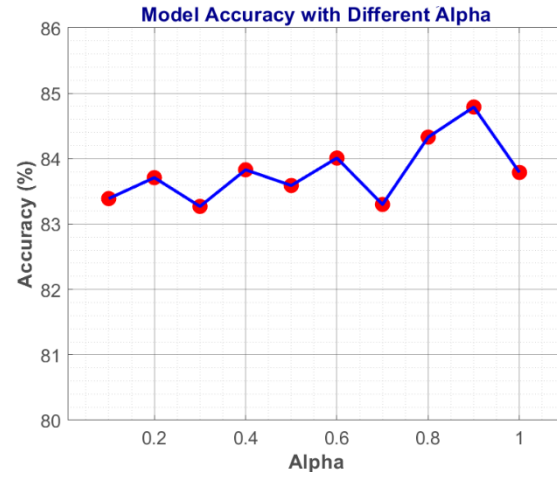
Table 5. Results for 5 folds of electrode 20 using knowledge distillation (T7-FT9)

Fold Number	Accuracy	Sensitivity	Specificity	F1-score	FPR
Fold1	84.60%	85.21%	84.04%	83.74%	0.1592
Fold2	85.0%	90.04%	80.96%	84.23%	0.1904
Fold3	84.51%	86.37%	82.95%	83.56%	0.1705
Fold4	84.55%	84.45%	84.63%	83.44%	0.1537
Fold5	83.83%	88.18%	80.33%	82.96%	0.1967
Average	84.49%	86.85%	82.58%	83.58%	0.1741

Table 6. Results for 5 folds of electrode 22 using knowledge distillation (FT10-T8)

Fold Number	Accuracy	Sensitivity	Specificity	F1-score	FPR
Fold1	83.94%	85.14%	82.89%	83.14%	0.1711
Fold2	85.53%	87.52%	83.93%	84.33%	0.1607
Fold3	84.22%	86.57%	82.26%	83.33%	0.1774
Fold4	84.06%	84.62%	83.59%	83.03%	0.1641
Fold5	84.05%	88.35%	80.58%	83.18%	0.1942

Average	84.36%	86.45%	82.65%	83.40%	0.173 5
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(a) (b)

Figure 11. The effect of parameters T and α on the knowledge distillation model, (a) Temperature, (b) Alpha.

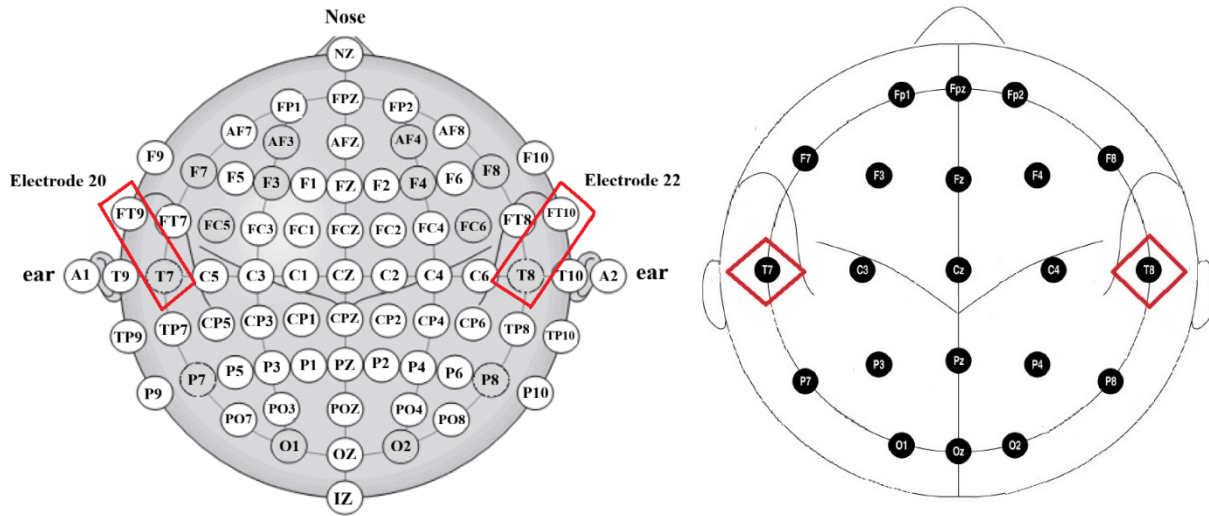


Figure 12. the selected electrodes in the international 10-20 System (right) and in the 10-10 system (left).

Table 7. Comparison of Parameters Between the Teacher and Student Models

Model	Parameters No.	Average Training Time (sec)	Average Testing Time (sec)
Teacher	560,068	16192,46	5.4
Student	334,212	5749,73	1

According to the comparison Table 8, we can observe that our method outperforms other existing methods in terms of accuracy, sensitivity, specificity, and F1 score. Also, the obtained p-value from statistical t-test show that the performance improvement of the proposed method is meaningful ($p\text{-value} < 0.05$). Although Wu et al. [44] reported higher evaluation metrics than our method, it is important to note that they used data from only 11 patients, while we utilized data from all 24 patients in the dataset to achieve our results. Naturally, by excluding patients whose seizures are more difficult to diagnose, the results can improve. In general, higher performance has been reported in articles that used a smaller number of patients.

In Table 8, the methods and types of preprocessing in related works and the proposed method are also shown. In recent studies, convolutional networks have demonstrated better performance when applied as patient-independent models. Additionally, due to the high importance of frequency bands in brain signals, the use of Mel-Spectrogram as a preprocessing technique and analyzing the brain signal in the frequency domain has significantly improved the model's performance. In the proposed method, seizures are predicted using only a single electrode, which requires less computational power and is more convenience for patients.

Table 8. Comparison of Related Works: Patient-Independent Models Using the Dataset CHB-MIT

Authors	Method	Preprocessing / Feature Extraction	Accuracy	Sensitivity	Specificity	F1-Score	FPR (h^{-1})	Prediction time (min)	No. of patient	No. of Channel	P-value
Affes et al.[11] 2022	SVM	Wavelet Transform/ BPF, EMD, MEMD	71.91%	78.9%	-	76.13 %	0.353	5	24	2	0.007
Cho et al.[48] 2017	SVM	Wavelet Transform/ BPF, EMD, MEMD	-	80.54%	80.5%	-	-	5	21	-	0.007
Khan et al.[43] 2017	2D-CNN	Band pass filter/ Wavelet Transform	-	87.8%	-	-	0.142	10	16	22	0.033
Hu et al.[47] 2019	CNN-SVM	Wavelet Transform PCA, HOG/ Mean amplitude, spectrum(MAS)	86.25%	81.20%	-	-	-	60	24	18	0.0045
Peng et al.[49] 2021	MMD-AAE	STFT power line noise removal	-	73%	-	-	0.24	30	16	23	0.012
Peng et al.[50] 2022	MAAE	STFT power line noise removal	-	82%	-	-	0.13	30	10	23	0.020
Wu et al.[44] 2023	2D-CNN	Signal compression	92.5%	94.2%	-	-	0.09	30	11	16	0.054
Zhang et al.[57] 2025	2D-CNN	STFT	-	85.0%	-	-	0.224	30	13	18	0.013

Our Proposed Network Teacher	3D-CNN	Mel-spectrogram Band pass Band stop Filter	87.52%	88.82%	86.36%	86.56 %	0.13 63	30	24	22	
Our Proposed Network KD-channel20	3D-CNN +KD	Mel-spectrogram Band pass Band stop Filter	84.49%	86.85%	82.58%	83.58 %	0.17 41	30	24	1	
Our Proposed Network KD-channel22	3D-CNN +KD	Mel-spectrogram Band pass Band stop Filter	84.36%	86.45%	82.65%	83.40 %	0.17 35	30	24	1	

V. CONCLUSION

This study aimed to predict the onset of seizures in patients with epilepsy, to reduce the complications associated with seizures by providing early predictions. For seizure prediction, deep learning-based models and knowledge distillation techniques have been used for model compression in order to reduce both time and hardware costs. Initially, a teacher model was designed and trained, which has a larger and more complex network. For the teacher model, 3D convolutional networks with 5 layers were used, while the input being the Mel-spectrogram of all 22 electrodes. Using dataset CHB-MIT the teacher model achieved an accuracy of 87.52%, a sensitivity of 88.82%, a specificity of 86.36%, and an F1-score of 86.56%. After designing the teacher model, the knowledge distillation technique was employed. Using this approach, the execution time, computational load, and the number of parameters of the teacher network were reduced, resulting in a student network capable of seizure prediction using only a single electrode, with four convolutional layers and Mel-spectrogram input. Using the student model, we identified two electrodes with superior performance. Electrode 20 achieved an accuracy of 84.49%, sensitivity of 86.85%, specificity of 82.58%, and an F1-score of 83.58%. Also electrode 22 reached an accuracy of 84.36%, sensitivity of 86.45%, specificity of 82.65%, and an F1-score of 83.40%. The proposed method, leveraging modern deep learning architectures and the knowledge distillation technique, has demonstrated high performance in both the teacher and student models compared to existing approaches. The proposed model has demonstrated both high accuracy and reliability in seizure prediction by employing only one electrode, yielding promising results and highlighting its potential for deployment in the real-world settings. On the other hand, the reduction in computational requirements of the student model has made the practical deployment of this model on mobile devices feasible. Therefore, the proposed method has opened new horizons in seizure prediction (by only one electrode) to improve the quality of life for individuals with epilepsy.

For future work, the development of two electrodes corresponding to the temporal region, T7-FT9 and FT10-T8, is proposed for using in a wearable device for seizure prediction. These two electrodes have demonstrated the better performance in seizure prediction of different epilepsy types. To confirm this, it is necessary to investigate this issue clinically and physiologically in future work. As a result, a wearable device can be designed using two electrodes placed near the patient's ear without causing discomfort. This device, operating on low-power hardware such as a smartphone, can predict an epileptic seizure 30 minutes in advance and alert the patient with a warning to help them to prepare before the onset of the seizure. Also, the integration of online learning models into this tool and determining the best features for seizure analysis from several options such as power spectral bands, wavelet/Gabor coefficients, phase correlation coefficient, higher order statistics or local binary patterns should be investigated in the future work.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT author statement

Hana Nyamoradi: Conceptualization, Methodology, Software, Data curation, Writing-Original draft preparation, Visualization, Investigation, Validation.

Abdolhossein Fathi: Supervision, Conceptualization, Methodology, Writing-Reviewing and Editing.

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Data availability

The original datasets, *CHB-MIT Scalp EEG Database*, used during the current study are available in the PhysioNet repository, [<https://physionet.org/content/chbmit/1.0.0/>]. Also, all extracted data from the original dataset will be available by corresponding author on reasonable request.

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